Chapter 7: Photosynthesis

Section 7.1: An Overview

Photosynthesis is the process by which green plants, algae and some bacteria convert light energy into chemical energy stored in carbohydrates. It is the entry point for almost all energy in the biosphere, and the origin of the oxygen in the atmosphere.

The overall reaction is usually written as:

$$6CO_2 + 6H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2$$

This summary equation is convenient but misleading. It suggests a single step, when in fact photosynthesis proceeds in two linked stages that occur in different parts of the chloroplast and depend on each other. The first stage captures light and stores its energy in two carrier molecules. The second stage spends that energy to fix carbon dioxide into sugar.

A second common misunderstanding is that the oxygen released comes from carbon dioxide. It does not. Isotope labelling experiments in the 1940s showed that every oxygen atom released comes from water, not from CO_2.

Section 7.2: The Leaf as a Photosynthetic Organ

The leaf is adapted to collect light and exchange gases while limiting water loss. A broad, flat lamina presents a large surface area to incoming light, and leaves are typically arranged to minimise mutual shading.

In cross section, a leaf shows several distinct layers. The upper epidermis is transparent and covered by a waxy cuticle that reduces evaporation. Beneath it lies the palisade mesophyll, a layer of column shaped cells packed with chloroplasts; this is where most photosynthesis occurs. Below that is the spongy mesophyll, whose loose arrangement creates air spaces allowing carbon dioxide to diffuse to the photosynthesising cells.

The lower epidermis contains stomata, pores flanked by guard cells. When the guard cells take up water they become turgid and bow apart, opening the pore. Stomata therefore control a trade off: opening them admits carbon dioxide but also permits water vapour to escape.

Section 7.3: Chloroplasts and Photosynthetic Pigments

The chloroplast is bounded by a double membrane. Inside, a system of flattened membranous sacs called thylakoids is arranged in stacks known as grana. The fluid surrounding the thylakoids is the stroma.

This structure matters because the two stages of photosynthesis are physically separated. The light dependent reactions take place on the thylakoid membranes, which hold the pigments and electron carriers. The light independent reactions take place in the stroma, which holds the enzymes of carbon fixation.

Chlorophyll a is the primary pigment and appears blue green. Chlorophyll b, along with accessory pigments such as carotenoids and xanthophylls, absorbs wavelengths that chlorophyll a does not and passes the energy on. Together the pigments absorb strongly in the blue and red regions of the spectrum and reflect green, which is why most foliage looks green to us.

Section 7.4: The Light Dependent Reactions

Pigments in the thylakoid membrane are organised into two photosystems, labelled Photosystem II and Photosystem I after the order of their discovery rather than the order in which they act.

When a photon strikes Photosystem II, energy is passed to a pair of chlorophyll a molecules at its reaction centre and an electron is raised to a higher energy level. This excited electron is captured by an acceptor and passed along a chain of carriers embedded in the membrane.

As electrons move down the chain they release energy, which is used to pump protons from the stroma into the thylakoid space. The resulting proton gradient drives ATP synthase, and ATP is produced. This mechanism is called chemiosmosis, and it is the same principle by which mitochondria make ATP during respiration.

Section 7.4 continued: Photolysis and the Fate of Electrons

Photosystem II has lost an electron and must replace it. It does so by splitting water, a reaction called photolysis:

$$2H_2O \rightarrow 4H^+ + 4e^- + O_2$$

The electrons replace those lost from the reaction centre, the protons contribute to the gradient across the thylakoid membrane, and the oxygen is released as a by-product. This is the source of atmospheric oxygen.

Meanwhile, electrons arriving at Photosystem I are re-energised by a second absorption of light and passed to $NADP^+$, reducing it to NADPH.

The light dependent stage therefore produces three things: ATP, NADPH, and oxygen. The first two are passed to the stroma; the third diffuses out of the leaf. Note that neither ATP nor NADPH is a long term store. Both are consumed within seconds by the reactions of the next stage.

Section 7.5: The Calvin Cycle

The light independent reactions occur in the stroma and are known as the Calvin cycle. They do not require light directly, but they depend entirely on the ATP and NADPH produced by the light dependent stage, so they stop shortly after darkness falls. Calling them the dark reactions is therefore misleading.

The cycle has three phases. In carbon fixation, a molecule of carbon dioxide is joined to the five carbon acceptor ribulose bisphosphate, or RuBP. This is catalysed by the enzyme RuBisCO, generally held to be the most abundant protein on Earth. The resulting six carbon compound is unstable and splits immediately into two molecules of the three carbon glycerate 3 phosphate.

In the reduction phase, ATP and NADPH from the light stage convert glycerate 3 phosphate into triose phosphate, a three carbon sugar.

Section 7.5 continued: Regeneration and Yield

Most of the triose phosphate produced does not leave the cycle. Five out of every six molecules are used, with further ATP, to regenerate RuBP so that the cycle can continue. Only one molecule in six is available for the synthesis of glucose, starch, cellulose, amino acids and lipids.

This accounting explains why the cycle must turn six times to yield a single molecule of glucose, consuming eighteen molecules of ATP and twelve of NADPH in the process.

RuBisCO is also inefficient in an important way. It can bind oxygen as well as carbon dioxide. When it binds oxygen the plant enters photorespiration, a pathway that consumes energy and releases previously fixed carbon without producing sugar. Photorespiration becomes more likely at high temperature and when stomata close, which restricts the supply of carbon dioxide.

Section 7.6: Factors Affecting the Rate

Three environmental factors principally determine the rate of photosynthesis: light intensity, carbon dioxide concentration and temperature.

At low light, the rate rises in direct proportion to intensity. As intensity increases further the curve flattens, because some other factor has become limiting. This illustrates the principle of limiting factors: when a process depends on several conditions, its rate is set by whichever is in shortest supply.

Carbon dioxide behaves similarly. Atmospheric concentration is around 0.04 per cent, which is low enough that CO_2 is frequently the limiting factor for crops in bright conditions. Commercial glasshouses often enrich the air to raise yields.

Temperature acts differently. Because the Calvin cycle is enzyme catalysed, the rate rises with temperature up to an optimum and then falls sharply as enzymes denature. A plant can therefore be damaged by heat even in ideal light.

Section 7.7: Adaptations - C4 and CAM Plants

Photorespiration wastes energy, and it is worst in hot, dry, bright conditions. Two groups of plants have evolved ways around it.

C4 plants, including maize and sugarcane, separate carbon capture from the Calvin cycle in space. Carbon dioxide is first fixed in mesophyll cells by the enzyme PEP carboxylase, which unlike RuBisCO has no affinity for oxygen. The four carbon product is then transported into bundle sheath cells, where it releases CO_2 at high concentration around RuBisCO. Photorespiration is suppressed.

CAM plants, such as cacti and pineapples, separate the two stages in time instead. They open their stomata at night, when evaporation is low, and fix carbon dioxide into organic acids stored in the vacuole. During the day the stomata close and the acids release CO_2 internally for the Calvin cycle. This conserves water at the cost of a slower overall rate.

Chapter Summary and Key Terms

Photosynthesis converts light energy into chemical energy in two linked stages. The light dependent reactions, on the thylakoid membranes, use light to produce ATP and NADPH and release oxygen from the photolysis of water. The Calvin cycle, in the stroma, spends that ATP and NADPH to fix carbon dioxide into triose phosphate.

Key terms: thylakoid, granum, stroma, photosystem, reaction centre, chemiosmosis, photolysis, NADPH, RuBisCO, ribulose bisphosphate, glycerate 3 phosphate, triose phosphate, photorespiration, limiting factor, C4, CAM.

Common misconceptions to check: the oxygen released comes from water and not from carbon dioxide; the light independent reactions are not the reactions that happen at night; and a plant in bright light is not necessarily photosynthesising faster, because some other factor may be limiting.